

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459898.85795 +/- 0.00081 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.03
Transit depth (Rp/Rs)^2	2.01 +/- 0.85 [%]
Orbital Inclination (inc)	85.3 +/- 1.45
Airmass coefficient 1 (a1)	1.007 +/- 0.024
Airmass coefficient 2 (a2)	-0.0029 +/- 0.0067
Scatter in the residuals of the lightcurve fit is	2.4 %
Best Comparison Star	#1 - [368, 302]
Optimal Aperture	4.82
Optimal Annulus	11.64
Transit Duration (day)	0.0835 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.03 vs 0.1301 (deviation 0.4σ)
Duration fit vs published	0.0835 d vs 0.0908 d (deviation 1.22σ)
Mid-time O–C	2.4 min
Depth SNR	4.7
Residual scatter	2.4 %

Light curve

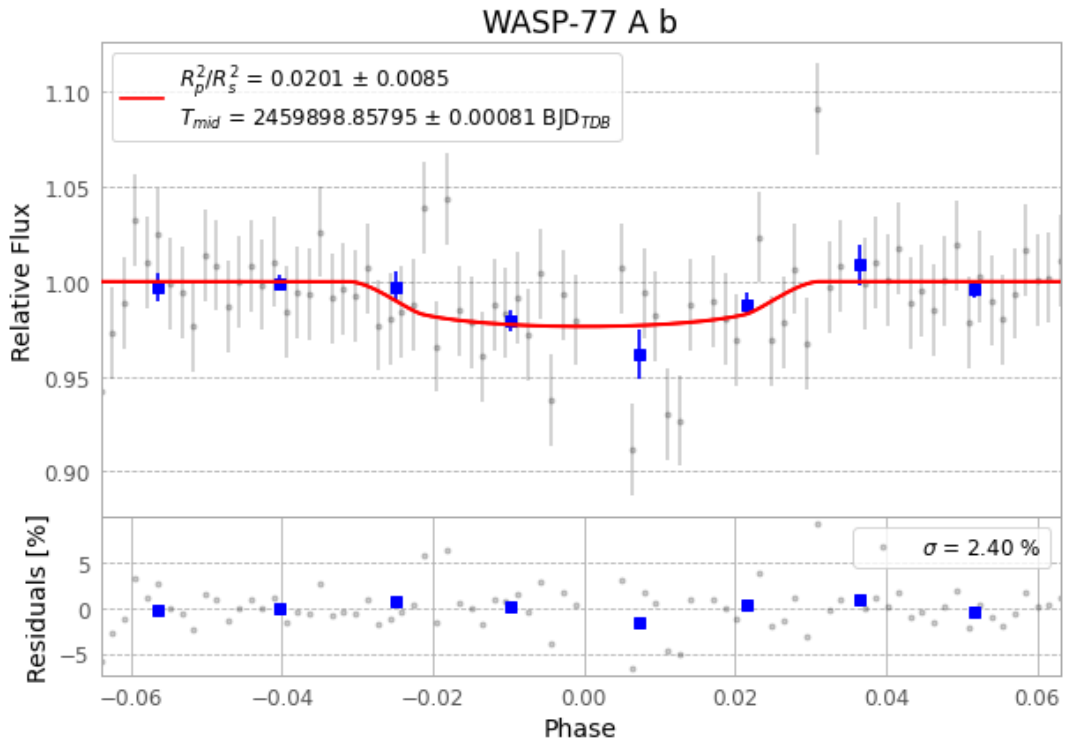


Figure 1 — FinalLightCurve_WASP-77 A b_2022-11-14.png (SHA-256 a91ab47ee97bb24f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [475, 187], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 95ef6e5e9508d622...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

84 FITS frames (55 MB), WASP-77221115062116.FITS → WASP-77221115103015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-221115.log (19fff747269b...), FinalLightCurve_WASP-77 A b_2022-11-14.png (a91ab47ee97b...), FinalLightCurve_WASP-77 A b_2022-11-14.csv (3699c256795b...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 10:35Z.